BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 40

Note: This section consists of 40 part questions and all are to be answered. Each question carries 01 marks. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 100 (Regular & Private)

Q1. Choose the correct answer for each from the given options:

Q1. (A')' = :

(A) B'
(B) U
(C) U'
(D) A

Q2. √2 is:

(A) Rational number
(B) Irrational number
(C) Odd number
(D) Even number

Q3. The value of i^9 is equal to:

(A) 1
(B) −i
(C) i
(D) −1

Q4. If z = 3 + 4i then z + $\bar{z}$ = :

(A) 6
(B) 8i
(C) 0
(D) 16

Q5. $2x^2 + 3y^2$ = :

(A) (2x + 3iy)(2x − 3iy)
(B) (√2x + √3iy)(√2x − √3iy)
(C) (2x − 3y)(2x + 3y)
(D) (√2x + √3y)(√2x − √3y)

Q6. The roots of quadratic equation $x^2 - 4x = 0$ are:

(A) Imaginary
(B) Rational and Different
(C) Irrational
(D) Rational and Equal

Q7. If D>0 and perfect square then roots of $ax^2 + bx + c = 0$ are:

(A) Equal
(B) Imaginary
(C) Rational
(D) Irrational

Q8. The sum of the all cube roots of unity is:

(A) 1
(B) 0
(C) w
(D) -1

Q9. If 4 and -5 are the roots of an equation then quadratic equation will be:

(A) $x^2 + x - 20 = 0$
(B) $x^2 - x + 20 = 0$
(C) $x^2 + x + 20 = 0$
(D) $x^2 - x - 20 = 0$

Q10. The matrix [[4,0,0],[0,4,0],[0,0,4]] is a:

(A) Diagonal matrix
(B) Scalar matrix
(C) Unit matrix
(D) Null matrix

Q11. If A = [[a,b],[c,d]] then Adj. A = :

(A) [[d,−b],[−c,a]]
(B) [[a,−b],[−c,d]]
(C) [[−a,c],[b,−d]]
(D) [[a,c],[b,d]]

Q12. [0] is:

(A) Square matrix
(B) Rectangular matrix
(C) Scalar matrix
(D) Unit matrix

Q13. If [[6,λ],[3,2]] is singular matrix then λ = :

(A) 4
(B) -4
(C) 12
(D) 18

Q14. For any two non-singular n-square matrices A and B, $(AB)^{-1}$ = :

(A) AB
(B) $B^{-1}A^{-1}$
(C) $A^{-1}B^{-1}$
(D) $A^{-1}B$

Q15. A binary operation * on a set is said to be associative if:

(A) a*b = b × a
(B) (a*b)*c = a*(b*c)
(C) a*b = b*a
(D) a*e = e*a = a

Q16. If first term is a and second term is b of H.P then d = :

(A) $(a-b)/ab$ (B) $ab/(a-b)$

(C) $(a+b)/(a-b)$ (D) $ab/(a+b)$

Q17. If $n \in N$, then number of terms in $(a + b)^n$ is:

(A) n (B) $(n - 1)$

(C) $(n + 1)$ (D) $(n + 2)$

Q18. The G.M between $\sqrt{2}$ and $1/\sqrt{2}$ is:

(A) ± 2 (B) $\pm\sqrt{2}$

(C) ± 1 (D) $\pm 1/\sqrt{2}$

Q19. The A.M. between $3\sqrt{5}$ and $5\sqrt{5}$ is:

(A) $\sqrt{5}$ (B) $2\sqrt{5}$

(C) $3\sqrt{5}$ (D) $4\sqrt{5}$

Q20. 0.97777 ..... =

(A) 0.97 (B) 0.97

(C) 0.97 (D) 0.97

Q21. The harmonic mean between 2a and 2b is:

(A) $\pm\sqrt{ab}$ (B) $(a+b)/2$

(C) $4ab/(a+b)$ (D) $2ab/(a-b)$

Q22. A coin is tossed twice, the probability of getting both tail is:

(A) 1/2 (B) 3/4

(C) 1/4 (D) 1

Q23. The value of ${}^nP_0$ is:

(A) 0 (B) 1

(C) n! (D) 1/n

Q24. The number of possible permutations of the letters of word EQUATION is:

(A) 40320 (B) 40032

(C) 40023 (D) 40230

Q25. The value of $(n+1)!/(n-1)!$ is:

(A) $n(n + 1)$ (B) $(n + 1)!$

(C) $(n+1)/(n-1)$ (D) $n(n + 1)!$

Q26. The middle term in the expansion $(1 + 2x)^6$ is:

(A) 1st term
(B) 4th term
(C) 2nd term
(D) 3rd term

Q27. If x is so small that its square and higher powers be neglected then $(1 - x)^{-2}$ = :

(A) $1 + 2x$
(B) $1 - 2x$
(C) $1 - x$
(D) $1 + x$

Q28. $\cos^2\theta - \sin^2\theta$ = :

(A) $\cos(\theta/2)$
(B) $\cos\theta$
(C) $\cos 2\theta$
(D) $-\cos 2\theta$

Q29. If in ΔABC a = 2cm, b = 3cm and c = 4cm then 2s = :

(A) 6cm
(B) 7cm
(C) 9cm
(D) 10cm

Q30. $\tan\theta \cot\theta$ = :

(A) $\cos\theta$
(B) $\sin\theta$
(C) $\sec\theta$
(D) $\operatorname{cosec}\theta$

Q31. If $\cos\theta > 0$ and $\sin\theta < 0$, then $\rho(\theta)$ lies in this quadrant:

(A) 1st
(B) 2nd
(C) 3rd
(D) 4th

Q32. $\sin(180° + \theta)$ = :

(A) $-\cos\theta$
(B) $-\sin\theta$
(C) $-\tan\theta$
(D) $\cot\theta$

Q33. $1 - \cos\theta$ is:

(A) $2\cos^2(\theta/2)$
(B) $2\tan^2(\theta/2)$
(C) $2\sec^2(\theta/2)$
(D) $2\sin^2(\theta/2)$

Q34. One degree in radians is equal to:

(A) $180/\pi$
(B) $3\pi/180$
(C) $\pi/180$
(D) one

Q35. $(\sin^2(\pi/2) + \cos^2(\pi/2)) - (\sin^2(\pi/3) + \cos^2(\pi/3))$ = :

(A) 1
(B) 0
(C) -1
(D) ∞

Q36. $\cos(-\theta) =$:

(A) $\sin\theta$ (B) $-\sin\theta$

(C) $\cos\theta$ (D) $-\cos\theta$

Q37. In the 3rd quadrant $\operatorname{cosec}\theta$ is:

(A) positive (B) negative

(C) both positive & negative (D) 0

Q38. The function $y = \sin^{-1}x$, its domain is:

(A) $-1 \le x \le 1$ (B) $-1 < x < 1$

(C) $0 \le x \le 1$ (D) $0 \le x < 1$

Q39. If the radius of the circle is 'r' then its area = :

(A) $2\pi r$ (B) πr

(C) πr^2 (D) $3\pi r$

Q40. In ΔABC, $a + b - c =$:

(A) $2s$ (B) $2(s - a)$

(C) $2(s - b)$ (D) $2(s - c)$

Section 'B' (Short Answer Questions)

Maximum Marks: 30

Note: Attempt any Ten part questions from this Section. All questions carry equal marks.

Q2.

(i) Solve the complex equation $(x + 3i)^2 = 2yi$.

(ii) Prove that $((1+\sqrt{-3})/2)^7 + ((-1-\sqrt{-3})/2)^7 + 1 = 0$.

(iii) $\sqrt{((t+16)/t)} + \sqrt{(t/(t+16))} = 2\ 1/12$

(iii-OR) If a, b are the roots of $ax^2 + bx + c = 0$, $a \neq 0$ find the equation whose roots are $1/\alpha$, $1/\beta$.

(iv) Show by using the properties of determinant: $|1\ 1\ 1;\ \alpha\ \beta\ \gamma;\ \beta\gamma\ \gamma\alpha\ \alpha\beta| = (\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$.

(v) Solve for x: $[[-2,3],[4,-1]]\ [[1,x,5],[2,4,x]]\ [[-3],[1],[0]] = [2\ -14]^t$

(vi) Show that multiplication is a binary operation on $S = \{1, -1\}$ but not on $T = \{1, 2\}$.

(vii) Which term of the sequence $-3, 3, 9, 15, \ldots\ldots$ is $6m - 15$?

(viii) Find the sum of n terms of a G.P. whose nth term is 3(3)^{n−1}

(ix) If a die is rolled twice, what is the probability that the sum of the points is 7?

(x) Prove the proposition by mathematical induction: $1 + 5 + 9 + \ldots\ldots + (4n - 3) = n(2n - 1)$

(x-OR) Find the term independent of x in the expansion of $(x - 2/x)^{10}$.

(xi) How far does a boy on a bicycle travel in 10 revolutions if the diameters of the wheels of his bicycle each equal to 56cm?

(xii) $\operatorname{cosec}\theta = -3/2$ and $\rho(\theta)$ is not in the fourth quadrant then find the remaining trinometric functions, by using radian function.

(xiii) Prove any two of the following: (a) $(\tan\theta + \sin\theta)/(\operatorname{cosec}\theta + \cot\theta) = \tan\theta \sin\theta$ ($\cos\theta \neq 0, -1$) (b) $\tan 3\theta = (3\tan\theta - \tan^3\theta)/(1 - 3\tan^2\theta)$ (c) $(\sin(\theta/2) - \cos(\theta/2))^2 = 1 - \sin\theta$

(xiv) Find the area of the triangle ABC, if $\alpha = 30°$, $\beta = 50°$, c = 90cm.

(xiv-OR) Draw the graph of the function cosq, where $-\pi < \theta < \pi$.

(xv) Solve the trionometric equation $\sqrt{3}\cos\theta + \sin\theta - 2 = 0$ or $\sin^{-1}\theta = \cos^{-1}\sqrt{(1-\theta^2)}$

Section 'C' (Detailed Answer Questions)

Maximum Marks: 20

> Note: Attempt any TWO questions from this Section.

Q3.

(a) Solve the equation: $1 + t - 4t^2 + t^3 + t^4 = 0$

(b) Apply Cramer's rule to solve the system of equations: $x + 2y + z = 8$; $2x - y + z = 3$; $x + y - z = 0$

Q4.

(a) Find the value of n so that $(a^{n+1} + b^{n+1})/(a^n + b^n)$ may become the G.M. between a and b, ($a \neq b$).

(b) If 'c' be a quantity, so small that c^3 may be neglected in comparison with 13, prove that: [illegible in source scan]

Q5.

(a) The base of a right angled triangle is 9cm and the sides of the triangle are in A.P. Find the hypotenuse.

(b) Show that in ΔABC : $r_1 = 4R \sin(\alpha/2)\cos(\beta/2)\cos(\gamma/2)$

(b-OR) Solve the equations: $x + y = 5$; $x^2 + 2y^2 = 17$